

DISTIL: Demonstration Distillation for Sample-Efficient Imitation Learning

Anonymous submission

Abstract

Expert demonstrations are the dominant cost of imitation learning, yet interactive methods spend them reflexively: a per-state gate fires on the first uncertain step of a rollout and the expert corrects that miss, whatever it is. Under a fixed budget, what matters is *which* failures get corrected and *how* each corrective demonstration is placed. We present **DISTIL**, an interactive imitation-learning loop that, out of the pool of all possible corrective demonstrations, distills the few most informative ones, the purest corrections, and spends the fixed budget only on those. DISTIL flags uncertain rollouts by the learner’s own per-step training loss, describes each flagged failure with a vision language model, performs root-cause reasoning grounded in a task knowledge graph, clusters failures into recurring modes with a cross-round memory, and lets a language model prescribe the round’s one corrective demonstration (a targeted correction of a recorded failure or a bridging demonstration covering several related ones) behind a feasibility check with re-prescription. The learner is any policy f_θ with a per-step loss: we instantiate pure CNN and MLP classifiers on a GridWorld task and state- and image-based diffusion policies on four robot manipulation tasks. Under an identical 20-demonstration budget against six interactive baselines, DISTIL attains the highest mean final success rate in all ten task-modality settings (e.g., 96.1% vs. 90.7% for Diff-Dagger on state Push-T; 95.3% vs. 89.6% on image Wipe), collects demonstrations with the highest per-demonstration information gain in every setting, and reports prescription confidences that correlate with realized policy improvement ($r = 0.82\text{--}0.89$).

Introduction

Behavior cloning regresses an expert’s actions on the states the expert visits (Pomerleau 1988, 1991), and its well-known failure mode is covariate shift: the learner’s own actions steer it off the demonstrated support, where errors compound over the horizon (Ross, Gordon, and Bagnell 2011). Interactive imitation learning fixes the distribution by aggregating expert corrections on states the *learner* visits (Ross, Gordon, and Bagnell 2011; Celemin et al. 2022), but the fix is paid for in expert effort: every correction is a demonstration

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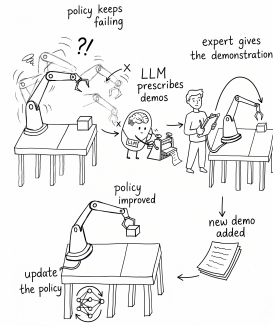


Figure 1: **DISTIL in a nutshell.** An LLM diagnoses the learner’s failures and *prescribes* the one corrective demonstration the expert should provide; the demonstration is added and the policy retrained. Out of all possible corrective demonstrations, DISTIL distills the few most informative ones and spends the fixed budget only on those.

an expert must produce. The query-efficient DAgger family therefore gates the expert behind a per-state predicate: SafeDAgger on action discrepancy (Zhang and Cho 2017), DropoutDAgger and EnsembleDAgger on predictive uncertainty (Menda, Driggs-Campbell, and Kochenderfer 2017, 2019), ThriftyDAgger on novelty and risk (Hoque et al. 2021a), and Diff-DAgger on the diffusion policy’s own training loss (Lee, Kang, and Kuo 2025). All of these decide *when* to interrupt a rollout; the expert then corrects whatever failure happened to trip the gate.

This is wasteful under the constraint that binds in practice: a *fixed demonstration budget*. When only twenty corrective demonstrations can be collected, the first uncertain step of the current rollout is rarely the best place to spend one. Failures repeat: a when-to-query gate can correct the same recurring mode again and again while rarer modes starve. Placement matters too: one demonstration started *between* several related failures can cover all of them, while an on-policy correction covers only its own rollout. Under a fixed budget the questions are *which* failures get corrected and *how* each demonstration is placed.

DISTIL (demonstration distillation) answers both questions with one loop (Figures 1 and 2). Rollouts of the current policy are monitored with the learner’s own per-step training loss, and uncertain episodes are flagged. A vision language model describes each flagged failure from its start, highest-loss (t^*), and end frames, and a reasoning model assigns a root cause grounded in a task knowledge graph. The failures are clustered into k recurring modes; a cluster memory carried across rounds rotates attention away from modes already corrected. A language model then *prescribes* the round’s single corrective demonstration: a targeted on-policy correction of one recorded failure, or a bridging demonstration placed at a new start state covering several related failures. A feasibility check with re-prescription and a deterministic fallback vets every prescription, so no budget is ever wasted. The expert supplies exactly the prescribed demonstration, it is added to the dataset, and the policy is updated; *which* mode to correct and *where* to start the demonstration are the two arms of the hybrid prescription, committed each round from the clustered evidence. The name is the mechanism: out of the pool of every corrective demonstration the expert could provide, DISTIL distills the few most informative ones, the purest corrections, and the fixed budget is spent only on those.

DISTIL is policy-agnostic. It needs only a learner f_θ with a per-step training loss ℓ_t that can be evaluated along rollouts: the cross-entropy loss of a pure CNN (image) or MLP (state) classifier on GridWorld, and the denoising loss of state- and image-based diffusion policies (Chi et al. 2023) on the robot tasks; the same loop runs unchanged over both families.

We evaluate on five tasks (GridWorld 5×5 , Push-T, Lift, Wipe, and Door), each in a state and an image modality, against six interactive baselines under an identical budget of 20 expert demonstrations. DISTIL attains the highest mean final held-out success rate in all ten task–modality settings, its demonstrations carry the highest per-demonstration information gain in every setting, and its prescription confidence predicts the realized change in success rate with Pearson $r = 0.82\text{--}0.89$.

Contributions. (1) We reframe budget-constrained interactive imitation learning as demonstration distillation: decide *which* failure modes to correct and *how* to place each corrective demonstration, rather than only *when* to interrupt a rollout. (2) DISTIL, a policy-agnostic loop combining uncertainty flagging, VLM failure perception, knowledge-grounded root-cause reasoning, failure-mode clustering with cross-round cluster memory, and LLM demonstration prescription with a feasibility-checked re-prescription loop. (3) A controlled study on five tasks $\times$ two modalities against six baselines: DISTIL has the best mean in all ten settings at the 20-demonstration budget, the highest per-demonstration information gain everywhere, and exposes a calibrated confidence signal ($r = 0.82\text{--}0.89$). (4) A bridge between online and offline imitation learning: DISTIL’s feasibility-checked prescriptions are standing artifacts, so the loop can pause after its on-policy analysis and the expert can record the prescribed demonstrations in one batch days later; expert time is decoupled from policy training time.

Related Work

Interactive Imitation Learning

Behavior cloning dates to ALVINN (Pomerleau 1988, 1991). DAgger aggregates expert labels on learner-visited states as no-regret online learning (Ross, Gordon, and Bagnell 2011); AggreVaTe and Deeply AggreVaTeD extend the reduction to cost-to-go (Ross and Bagnell 2014; Sun et al. 2017); Celemin et al. (2022) survey the field. Because vanilla DAgger queries constantly, query-efficient variants gate *when* to defer: SafeDAgger learns a discrepancy classifier (Zhang and Cho 2017); DropoutDAgger and EnsembleDAgger threshold dropout and ensemble uncertainty (Menda, Driggs-Campbell, and Kochenderfer 2017, 2019; Gal and Ghahramani 2016; Lakshminarayanan, Pritzel, and Blundell 2017); HG-DAgger and LazyDAgger manage human-gated control switching (Kelly et al. 2019; Hoque et al. 2021b); ThriftyDAgger budgets interventions by novelty and risk (Hoque et al. 2021a); DART injects noise into the expert’s rollouts so offline collection covers shifted states (Laskey et al. 2017); IWR and Sirius reweight samples around human takeovers (Mandlekar et al. 2020; Liu et al. 2023). Diffusion policies (Chi et al. 2023; Ho, Jain, and Abbeel 2020; Janner et al. 2022) capture the multimodal actions that make agreement-based uncertainty brittle, and Diff-DAgger queries on the diffusion training loss itself (Lee, Kang, and Kuo 2025). All decide when to interrupt; none reasons over the *set* of a policy’s failures, remembers what was corrected, or chooses where a demonstration should begin. That gap is DISTIL’s subject.

Foundation Models for Failure Reasoning

Language and vision-language models increasingly steer robot data collection and control: SayCan grounds plans in affordances (Ahn et al. 2022), Inner Monologue closes a textual feedback loop (Huang et al. 2022), and PaLM-E and RT-2 fuse perception with language (Driess et al. 2023; Brohan et al. 2023). A second line emits executable structure: policy code (Liang et al. 2023; Singh et al. 2023), 3D value maps (Huang et al. 2023), or reward functions (Ma et al. 2024). Closest to us is failure reasoning: REFLECT narrates and corrects robot failures (Liu, Bahety, and Song 2023), AHA reasons over manipulation failures (Duan et al. 2025), and Reflexion and Self-Refine iterate on model outputs (Shinn et al. 2023; Madaan et al. 2023). These systems reason about one trajectory at a time; DISTIL aggregates VLM-perceived failures across rollouts, clusters them into modes, and narrows the LLM’s output to one auditable artifact: a feasibility-checked prescription for a single corrective demonstration.

Active Selection, Representations, and Resets

Deciding which failures deserve a demonstration is batch active learning over failures: pool-based selection under a budget (Settles 2009), core-set/ k -center coverage (Sener and Savarese 2018), uncertainty–diversity hybrids (Ash et al. 2020), and farthest-point sampling (Eldar et al. 1997). Whereas these select unlabeled inputs, DISTIL selects failure

modes and adds a cross-round memory so coverage accumulates over the whole run. Image-based failures are embedded with a frozen R3M encoder (Nair et al. 2022). Bridging demonstrations placed at prescribed start states relate to reverse-curriculum reset generation (Florensa et al. 2017) and learning-to-reset (Eysenbach et al. 2018); DISTIL’s start states follow clustered failure evidence rather than a difficulty schedule.

Method

Setup: Budgeted Interactive Imitation

Each task is a finite-horizon decision process with states s , actions a (discrete moves on GridWorld; relative joint-position deltas on the robot tasks), and an expert π^* (a human on GridWorld; privileged controllers on the robot tasks). The learner is *any* policy f_θ trained on an aggregated demonstration dataset $\mathcal{D}$ by minimizing a per-step loss ℓ ,

$$\theta^* = \arg \min_{\theta} \mathbb{E}_{(s,a) \sim \mathcal{D}} [\ell(f_\theta; s, a)], \quad (1)$$

where ℓ is the binary cross-entropy over the expert’s action labels for the GridWorld classifiers (pure CNN on images, MLP on states) and the denoising (v -prediction) loss for the diffusion policies on the robot tasks (Ho, Jain, and Abbeel 2020; Chi et al. 2023). Evaluated along a rollout at the learner’s own executed actions, the loss yields a per-step uncertainty signal

$$\ell_t = \ell(f_\theta; s_t, a_t^{\text{nov}}), \quad (2)$$

which scores how far the visited pair sits from the training distribution (Lee, Kang, and Kuo 2025). Interactive learning proceeds in rounds: round r produces one corrective demonstration d_r (a segment of expert state–action pairs), and

$$\mathcal{D}_{r+1} = \mathcal{D}_r \cup d_r, \quad \theta_{r+1} = \arg \min_{\theta} \mathbb{E}_{\mathcal{D}_{r+1}} [\ell]. \quad (3)$$

All methods receive the same hard budget of $B=20$ corrective demonstrations, one per round; the only degree of freedom is how each demonstration is chosen.

Baseline Query Rules

Every baseline instantiates one template: a scalar score is computed at each visited state, and the expert takes over at the first threshold crossing,

$$\text{Query}_m(s_t) = \mathbf{1}[\text{score}_m(s_t) > \tau_m]. \quad (4)$$

SafeDagger scores the novice–expert action discrepancy $\|a_t^{\text{nov}} - a_t^{\text{exp}}\|_2$ ($\tau=0.1$; on GridWorld, the fraction of rollout steps whose action disagrees with the expert’s action labels) (Zhang and Cho 2017). DropoutDagger draws N stochastic action samples and fires when the fraction landing within a τ -ball of the expert action drops below p (Menda, Driggs-Campbell, and Kochenderfer 2017; Gal and Ghahramani 2016). EnsembleDagger fires on the doubt (variance) of $M=5$ independently trained members or on their mean discrepancy (Menda, Driggs-Campbell, and Kochenderfer 2019; Lakshminarayanan, Pritzel, and Blundell 2017). ThriftyDagger fires on novelty or on learned task risk $1 - Q_\psi(s, a)$, with quantile-calibrated thresholds (Hoque

et al. 2021a). Stagger is a uniform-random control that corrects one uniformly chosen recorded failure per round. On the robot tasks we also run Diff-Dagger (Lee, Kang, and Kuo 2025), whose score is the diffusion loss of Eq. 2 itself: with $\hat{F}$ the empirical CDF of training-set losses and $\alpha=0.99$,

$$\text{Query}_{\text{Diff}}(s_t) = \mathbf{1}[\hat{F}(\ell_u) > \alpha \quad \forall u \in (t-K, t)], \quad (5)$$

recalibrated at every retraining. The rosters differ for a structural reason: Diff-Dagger’s query rule *is* the diffusion policy’s own training loss, so it requires a diffusion learner and does not apply to the discrete-action CNN/MLP classifiers on GridWorld, where Stagger is the matched uniform-random control; conversely, Stagger’s random pick assumes any visited state can be re-instantiated as a demonstration start, which holds for grid layouts but not for contact-rich manipulation states, so on the robot tasks Diff-Dagger, the loss-native rule, is the control (six baselines overall, five per task). Whatever the score, each rule decides *when* to interrupt and corrects that rollout from the crossing step; none compares failures, remembers corrected modes, or chooses where the demonstration starts.

The DISTIL Loop

DISTIL replaces the per-state gate with a round-level analysis of the whole failure set, run identically for every learner: from the pool of every corrective demonstration the expert could give, it distills the few most informative ones and spends the budget only on those. Algorithm 1 gives the loop.

Uncertainty flagging and failure perception. Each round rolls the current policy and collects the failure set $\mathcal{F}_r = \{f_1, \dots, f_N\}$. Along each episode the per-step loss ℓ_t (Eq. 2) is logged; steps whose loss exceeds an adaptive threshold (the 95th percentile of the episode’s loss distribution) are flagged, and each failure retains its peak-loss step

$$t_i^* = \arg \max_t \ell_t^{(i)}, \quad \text{peak}_i = \max_t \ell_t^{(i)}, \quad (6)$$

taken as the failure point. A vision language model then describes three frames per failure (start, t_i^* , and end) and produces a textual failure report (Bai et al. 2025; Liu, Bahety, and Song 2023; Duan et al. 2025). A reasoning model (Yang et al. 2025) assigns each failure a root cause (e.g., wrong approach, no contact, wrong orientation, timeout) and a task phase. Both stages are grounded by knowledge-augmented generation (KAG): a per-task knowledge graph encoding the embodiment, object and goal geometry, workspace bounds, a failure taxonomy, and per-failure-mode prescription rules is rendered into the prompt.

Failure-mode clustering. Each failure is featurized by a descriptor anchored at t_i^* . For state-based robot runs this is the 6-D quaternion-free geometric vector

$$\phi_i = [p_{i,x}, p_{i,y}, \sin \theta_i, \cos \theta_i, \rho_i, \delta_i], \quad (7)$$

with $(p_{i,x}, p_{i,y}, \theta_i)$ the object’s planar pose, $\rho_i = t_i^*/T_i$ the task progress, and δ_i the end-effector contact distance; for image-based runs the descriptor is a frozen R3M embedding (Nair et al. 2022) of the same three frames, PCA-reduced

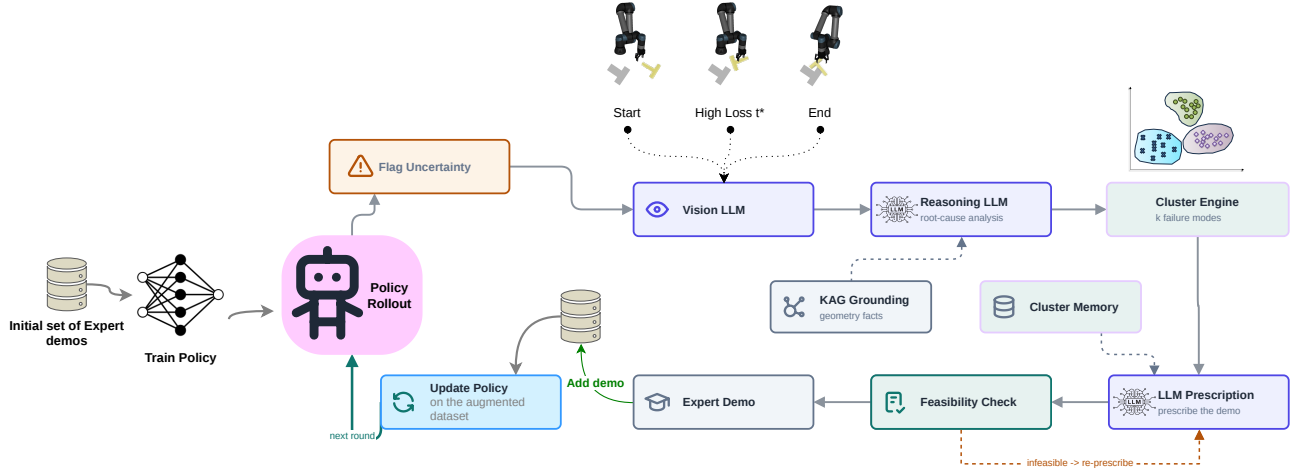


Figure 2: **The DISTIL loop.** Rollouts of f_θ are monitored for uncertainty via the per-step training loss; a vision LLM summarizes each flagged failure’s start, highest-loss (t^*), and end frames, and a reasoning LLM assigns root causes grounded in geometric facts (KAG); failures are clustered into k recurring modes with cross-round memory, the LLM prescribes the round’s corrective demonstration behind a feasibility check with re-prescription, and the accepted demonstration augments the dataset before the policy is updated.

to $k' = \min(16, N-1)$ dimensions. On GridWorld, ϕ_i uses the agent’s cell, its signed offset to the goal, progress, and Manhattan distance-to-goal. The standardized descriptors $\tilde{X}$ are clustered agglomeratively, with the number of modes chosen by maximum mean silhouette,

$$k^* = \arg \max_{k \in [2, k_{\max}]} \text{sil}(k), \quad (8)$$

with $k_{\max} = \max(2, \min(6, N-1))$. Each cluster C keeps its representative $\text{rep}(C) = \arg \min_{i \in C} \|\tilde{X}_i - \bar{X}_C\|_2$, centroid geometry computed on raw poses, and mean peak loss $\bar{L}_C$; the dominant cluster C^* is the largest, breaking ties by $\bar{L}_C$.

Cluster memory. Because the failure distribution shifts after every retraining, a persistent mode could absorb the entire budget. DISTIL therefore remembers previously corrected cluster centroids, penalizes recently covered regions with a recency-discounted Gaussian kernel, and rotates the target among clusters that rival the dominant one:

$$P_{\text{mem}}(c) = \sum_{(r_i, c_i) \in \text{Mem}} \gamma^{r-r_i} \exp\left(-\frac{\|c-c_i\|_2^2}{2\sigma^2}\right),$$

$$C_{\text{tgt}} = \arg \max_{C: |C| \geq |C^*| - 1} (\bar{L}_C - \lambda P_{\text{mem}}(c_C)), \quad (9)$$

with $\gamma=0.6$, $\sigma=0.06$, $\lambda=1$. The prescription is also conditioned on a small diverse context set S ($|S| \leq \kappa=3$): the target representative is forced in, the worst-peak-loss failure is seeded, and remaining slots are filled by farthest-point selection in descriptor space (Sener and Savarese 2018; Eldar et al. 1997).

Demonstration prescription. Given the clustered evidence, root causes, and cited context failures, the LLM prescribes the round’s single corrective demonstration together

with a self-reported confidence. The prescription is a hybrid decision with two arms, both always in play. *Targeted correction*: the LLM selects one recorded failure; its scene is re-instantiated, the current policy replays to its divergence point t^* , and the expert takes over from there. *Bridging place-ment*: the LLM instead places the demonstration at a new start state ξ positioned between two or three cited failures of the same mode, so one demonstration covers all of them. The bridge pose is anchored to the target representative A : the commanded offset is L_2 -capped in position, clamped in orientation, and projected onto the KAG workspace bounds $\mathcal{W}$,

$$\Delta_{xy} = \text{cmd}.p_{xy} - p_{A,xy}, \quad \Delta_{xy} \leftarrow \Delta_{xy} \min\left(1, \frac{\Delta_{\max}}{\|\Delta_{xy}\|_2}\right),$$

$$\Delta_\theta = \text{clamp}(\text{wrap}_\pi(\text{cmd}.\theta - \theta_A), \pm\theta_{\max}),$$

$$\xi = \text{clamp}_{\mathcal{W}}(p_A + [\Delta_{xy}, 0], \theta_A + \Delta_\theta, q_A), \quad (10)$$

with $(\Delta_{\max}, \theta_{\max}) = (0.06 \text{ m}, 0.4 \text{ rad})$ for bridges on Push-T. On Lift the cube start is clamped to $\Delta_{\max} = 0.03 \text{ m}$ in x and y with $\theta_{\max} = 0$ (no yaw randomization); on Door only the frame’s xy shifts, with $(\Delta_{\max,x}, \Delta_{\max,y}) = (0.0135, 0.013) \text{ m}$ and $\theta_{\max} = 0$ (the bridge reuses the representative failure’s orientation), and on Wipe the randomized quantity is a whole dirt-marker path, so bridging is infeasible and every prescription is a targeted correction. On GridWorld the prescription is a corridor layout (start, goal, obstacles, and a cell sequence) that the human expert is constrained to follow.

Feasibility check and re-prescription. Every prescription is validated before any expert effort is spent: robot prescriptions must respect $\mathcal{W}$ and cite real recorded failures; GridWorld corridors must stay on the grid, avoid the obstacles, and connect start to goal under an A^* /BFS path-validity

Algorithm 1 The DISTIL loop (one demonstration per round).

Input: initial demos $\mathcal{D}_0$, expert π^* , budget $B=20$

Output: trained policy f_θ

- 1: train f_θ on $\mathcal{D}_0$ (Eq. 1); $\text{Mem} \leftarrow \emptyset$
 - 2: **for** $r = 1$ **to** B **do**
 - 3: roll out f_θ ; flag uncertain episodes by ℓ_t ; record t_i^* , peak_i (Eqs. 2, 6); VLM describes start/ t_i^* /end frames; reasoning LLM assigns KAG-grounded root causes
 - 4: featurize (Eq. 7); cluster into k^* modes (Eq. 8); rotate target via cluster memory (Eq. 9); build context set S
 - 5: LLM prescribes the demonstration with confidence: targeted correction of a cited failure, or bridging start state ξ (Eq. 10); re-prescribe while infeasible or empty; after bounded attempts, fall back to the nearest untried failure
 - 6: expert provides d_r ; $\mathcal{D}_r \leftarrow \mathcal{D}_{r-1} \cup d_r$; append target centroid to Mem; update f_θ on $\mathcal{D}_r$ (Eq. 3)
 - 7: **end for**
 - 8: **return** f_θ
-

check. An infeasible or empty prescription is rejected and re-prescribed with the rejection fed back; after a bounded number of attempts the round falls back deterministically to a targeted correction of the nearest untried recorded failure, so a round always yields a valid demonstration. The accepted demonstration $d_r = \{(s_t, \pi^*(s_t)) : t \geq t^*\}$ is added, the target centroid is appended to the memory, and the policy is updated (Eq. 3).

Relation to the query rules. The baselines of Eqs. 4–5 are the degenerate round: a single-episode failure set, no clustering or memory, and the prescription fixed to the targeted arm at the gate’s own crossing step; DISTIL spends the identical budget against the whole failure distribution and the history of what was already corrected.

Bridging online and offline imitation. DAgger-family methods are *online* interactive imitation learning: an expert must be on call throughout training, because corrections are collected while the policy rolls out. Classical behavior cloning is *offline*: collect demonstrations, then train. DISTIL bridges the two. Its prescriptions are standing artifacts: the loop can run its rollout, clustering, and prescription analysis on-policy and then pause; the prescribed start configurations remain valid, and the expert can return days later to record the prescribed demonstrations in one batch before the loop resumes. Expert time is decoupled from policy training time: the expert schedules a recording session instead of supervising every rollout.

Experiments

We ask three questions. **(Q1)** Under an identical 20-demonstration budget, does DISTIL reach a higher final success rate than when-to-query baselines? **(Q2)** Does it collect more informative demonstrations, and how does information gain relate to allocation across failure modes? **(Q3)** Are

the LLM’s prescription confidences predictive of realized improvement, and are the discovered failure modes semantically meaningful?

Setup

Tasks and experts. **GridWorld** 5×5 : the agent moves from a start cell to a goal cell with four discrete moves on a 5×5 grid containing three obstacles; an episode succeeds when the agent reaches the goal. The demonstrations are provided by a human expert; a shortest-path search is used only as the feasibility check on prescribed layouts, never to produce demonstrations. **Push-T** (ManiSkill3 (Tao et al. 2024; Mu et al. 2021; Gu et al. 2023); task from Implicit BC (Florence et al. 2021)): a Franka Panda with a stick end-effector pushes a T-shaped block onto a fixed goal pose; the expert is a privileged PPO policy. **Lift**, **Wipe**, and **Door** (robosuite (Zhu et al. 2020), evaluation following robomimic (Mandlekar et al. 2021)): lift an object off the table, wipe a mark off the table, and open a door, each with a scripted motion-planner expert. Every task runs in a *state* modality (privileged proprioception and object state) and an *image* modality (RGB observations hiding object state).

Learners. On GridWorld the image learner is a pure CNN and the state learner an MLP, both trained with cross-entropy on the expert’s action labels. On all four robot tasks every method shares one diffusion-policy backbone (Chi et al. 2023; Ho, Jain, and Abbeel 2020), a conditional 1-D U-Net denoiser with a v -prediction objective, paired with an MLP state encoder or an end-to-end finetuned R3M ResNet-18 image encoder with spatial-softmax keypoints (Nair et al. 2022). Within each cell all methods share the identical learner and behavior-cloned initialization per seed; arms differ only in how demonstrations are acquired.

Protocol and metrics. Every method receives a budget of exactly **20 expert demonstrations**, added one per round (Eq. 3); the policy is retrained after each addition and evaluated on a fixed held-out set shared across methods and seeds. We run **9 seeds** per cell on GridWorld and **5 seeds** per cell on the robot tasks. The primary metric is the final held-out success rate (SR) at the budget. As analysis metrics we report the *per-demonstration information gain* (each round, before fine-tuning on the newly collected demonstration, the current policy’s per-step loss evaluated on that demonstration) and, for DISTIL, the Pearson correlation between prescription confidence and the realized change in success rate after the demonstration is incorporated.

Main Results (Q1)

Table 1 reports final held-out success rates in all ten cells; Figure 3 shows learning curves on three representative ones. **DISTIL attains the highest mean success rate in every cell.** On state Push-T it reaches 96.1% against 90.7% for Diff-DAgger, a +5.4-point margin on the task family Diff-DAgger was designed for. Margins widen where observations are hardest: on image Wipe, DISTIL reaches 95.3% vs. 89.6% for Diff-DAgger and 69.6% for SafeDAgger (+25.7 points), consistent with a gate that keeps re-correcting the same contact-rich mode while the uncertainty-gated curves

Table 1: **Final held-out success rate (%) at the 20-demonstration budget**, mean $\pm$ std over seeds (9 per cell on GridWorld, 5 on the robot tasks). Columns abbreviate SafeDagger, DropoutDagger, EnsembleDagger, ThriftyDagger; Stagger runs on GridWorld, Diff-Dagger on the robot tasks. Best mean per row in **bold**: DISTIL is highest in all ten settings; seed-level spreads overlap in the tighter cells (see text).

Task	Obs.	Safe	Dropout	Ensemble	Thrifty	Stagger	Diff-Dagger	DISTIL (Ours)
GridWorld 5 $\times$ 5	state	85.3 $\pm$ 2.7	84.9 $\pm$ 2.5	86.2 $\pm$ 2.1	86.8 $\pm$ 2.0	85.7 $\pm$ 1.5	–	89.9 $\pm$ 1.3
GridWorld 5 $\times$ 5	image	86.1 $\pm$ 2.8	85.8 $\pm$ 2.6	85.7 $\pm$ 2.2	87.1 $\pm$ 1.9	86.6 $\pm$ 2.3	–	89.6 $\pm$ 1.8
Push-T	state	82.0 $\pm$ 6.8	84.8 $\pm$ 6.1	85.9 $\pm$ 5.8	83.2 $\pm$ 7.2	–	90.7 $\pm$ 4.5	96.1 $\pm$ 4.5
Push-T	image	78.1 $\pm$ 7.8	82.1 $\pm$ 6.9	83.2 $\pm$ 6.6	79.3 $\pm$ 8.1	–	89.0 $\pm$ 4.8	93.9 $\pm$ 4.9
Lift	state	99.2 $\pm$ 1.6	99.2 $\pm$ 1.0	99.2 $\pm$ 1.0	98.8 $\pm$ 2.4	–	99.2 $\pm$ 1.0	100.0 $\pm$ 0.0
Lift	image	99.6 $\pm$ 0.8	97.2 $\pm$ 3.5	98.8 $\pm$ 1.6	99.6 $\pm$ 0.8	–	99.6 $\pm$ 0.8	100.0 $\pm$ 0.0
Wipe	state	88.0 $\pm$ 2.5	89.6 $\pm$ 4.1	90.8 $\pm$ 4.3	90.0 $\pm$ 2.5	–	90.4 $\pm$ 6.0	95.5 $\pm$ 6.0
Wipe	image	69.6 $\pm$ 5.3	83.2 $\pm$ 6.8	84.4 $\pm$ 7.1	69.2 $\pm$ 9.0	–	89.6 $\pm$ 3.2	95.3 $\pm$ 3.2
Door	state	93.2 $\pm$ 5.2	92.8 $\pm$ 2.7	88.8 $\pm$ 7.0	89.6 $\pm$ 3.9	–	95.2 $\pm$ 4.3	98.4 $\pm$ 4.2
Door	image	92.4 $\pm$ 3.2	88.8 $\pm$ 3.3	86.0 $\pm$ 10.9	92.8 $\pm$ 2.7	–	89.2 $\pm$ 3.5	99.2 $\pm$ 3.4

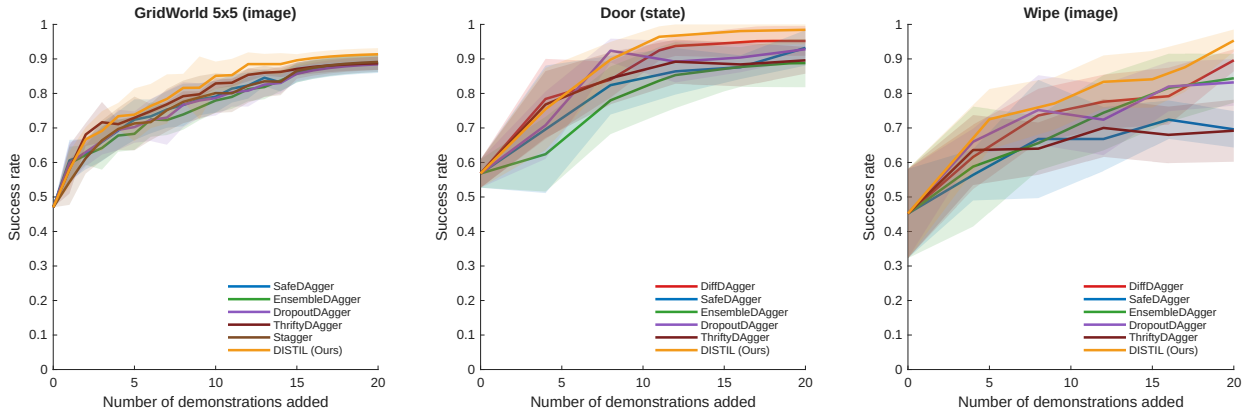


Figure 3: **Success rate vs. demonstrations added** on GridWorld 5 $\times$ 5 (image), Door (state), and Wipe (image); mean $\pm$ std over seeds (9 for GridWorld, 5 for the robot tasks). Under the same 20-demo budget, DISTIL dominates throughout training, with the largest margin on image-based Wipe, where uncertainty-gated baselines plateau.

Table 2: **Per-demonstration information gain** (pre-finetune policy loss on each newly acquired demonstration; suboptimal demonstrations are ruled out by construction, so higher = a demonstration from a region the policy has not learned), mean $\pm$ std pooled across a cell’s seeds. Abbreviations as in Table 1; D-Dag is Diff-Dagger. DISTIL is the most informative in all ten settings.

Task	Obs.	Safe	Dropout	Ensemble	Thrifty	Stagger	D-Dag	DISTIL
GridWorld state	2.55 $\pm$ 1.68	2.95 $\pm$ 2.12	1.33 $\pm$ 1.18	1.34 $\pm$ 0.93	1.83 $\pm$ 1.11	–	–	3.55 $\pm$ 2.51
GridWorld image	2.46 $\pm$ 1.61	2.53 $\pm$ 1.69	1.57 $\pm$ 1.55	1.37 $\pm$ 1.12	1.88 $\pm$ 1.43	–	–	3.21 $\pm$ 2.33
Push-T state	1.66 $\pm$ 0.99	2.36 $\pm$ 1.63	1.11 $\pm$ 0.64	1.10 $\pm$ 0.65	–	1.57 $\pm$ 1.10	2.81 $\pm$ 2.09	
Push-T image	2.04 $\pm$ 1.11	2.16 $\pm$ 1.36	1.06 $\pm$ 0.64	1.20 $\pm$ 0.62	–	1.80 $\pm$ 1.10	2.82 $\pm$ 1.72	
Lift state	2.23 $\pm$ 1.36	2.10 $\pm$ 1.25	1.12 $\pm$ 0.74	1.13 $\pm$ 0.55	–	1.61 $\pm$ 1.12	2.64 $\pm$ 1.65	
Lift image	2.18 $\pm$ 1.40	2.17 $\pm$ 1.39	1.00 $\pm$ 0.55	1.21 $\pm$ 0.64	–	1.36 $\pm$ 0.85	2.93 $\pm$ 1.67	
Wipe state	2.02 $\pm$ 0.96	2.38 $\pm$ 1.55	1.23 $\pm$ 1.10	1.18 $\pm$ 0.81	–	1.43 $\pm$ 0.80	2.91 $\pm$ 2.02	
Wipe image	2.50 $\pm$ 1.47	2.96 $\pm$ 2.00	1.43 $\pm$ 0.89	1.52 $\pm$ 0.86	–	1.95 $\pm$ 1.16	3.62 $\pm$ 2.19	
Door state	2.53 $\pm$ 1.64	3.10 $\pm$ 2.21	1.43 $\pm$ 0.82	1.42 $\pm$ 0.90	–	1.84 $\pm$ 1.11	3.43 $\pm$ 2.12	
Door image	2.35 $\pm$ 1.40	2.46 $\pm$ 1.44	1.24 $\pm$ 0.76	1.26 $\pm$ 0.72	–	1.58 $\pm$ 0.92	3.00 $\pm$ 1.98	

of Figure 3 flattened by mid-budget. On image Door, DISTIL reaches 99.2%, +6.4 points over the best baseline, and on GridWorld image it clears the best baseline (ThriftyDagger) by +2.5 points with the smallest variance in the column. Lift

is near-saturated for all methods, yet only DISTIL closes it completely (100.0 $\pm$ 0.0 in both modalities). With 5–9 seeds the seed-level spreads overlap the best baseline in the tighter cells (GridWorld, Lift, state Door), so we do not claim per-cell separation there; the evidence is the consistent ranking (best mean in all ten cells) plus the wide margins on Push-T, Wipe, and image Door.

Information Gain and Its Allocation (Q2)

Table 2 and Figure 4 quantify how much new information each acquired demonstration carries: each round, before fine-tuning on the newly collected demonstration, we evaluate the current policy’s per-step loss on it. A high pre-finetune loss can mean only two things: (a) the demonstration covers a region the current policy has not learned (diverse data from an underrepresented region), or (b) the demonstration is suboptimal or invalid. Case (b) is ruled out by construction: every prescription passes the feasibility check before any demonstration is collected, and the demonstrations themselves are produced by the expert. High pre-finetune loss

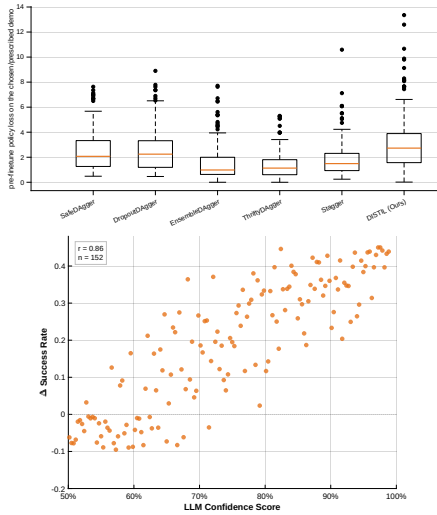


Figure 4: **Top: per-demonstration information gain** (Grid-World 5×5, image): the policy’s pre-finetune loss on each newly acquired demonstration; DISTIL’s prescribed demonstrations carry the most new information. **Bottom: prescription confidence predicts policy improvement** (same setting): each point is one prescribed demonstration ($r = 0.86$); across all ten task-modality settings, r ranges from 0.82 to 0.89.

therefore identifies demonstrations from underrepresented regions, and the per-demonstration information gain measures exactly the diversity a method’s chosen demonstrations add. DISTIL’s prescribed demonstrations have the highest mean information gain in all ten cells (e.g., 3.62 vs. next-best 2.96 on image Wipe; 3.55 vs. 2.95 on state GridWorld), with the widest upper spread in the boxplot: the prescription deliberately seeks out what the policy cannot yet do. One caveat: SafeDagger and DropoutDagger collect higher raw information gain than EnsembleDagger and ThriftyDagger, yet the information-gain orderings do not match the success-rate orderings of Table 1, so high per-demonstration gain is necessary but not sufficient. What converts it into success rate is its *allocation across failure modes*: a gate that repeatedly harvests high-loss demonstrations from the *same* mode accumulates redundant information, whereas DISTIL’s clustering and cluster memory spread high-loss demonstrations over distinct, not-yet-corrected modes; this reading matches the flattened baseline curves of Figure 3.

Confidence Calibration and Discovered Modes (Q3)

Each prescription carries a self-reported confidence. Why should the LLM’s prescription be trusted in each round? The success-rate feedback arrives late: only after the demonstration is collected, the policy retrained, and new rollouts evaluated. The prescription confidence is reported at prescription time, blind to that future outcome and to the success-rate trajectory. Figure 4 shows that this blind, self-reported confidence correlates with the realized policy improvement on image GridWorld ($r = 0.86$); the same computation per cell yields $r = 0.88$ (state GridWorld), $0.87/0.88$ (state/image

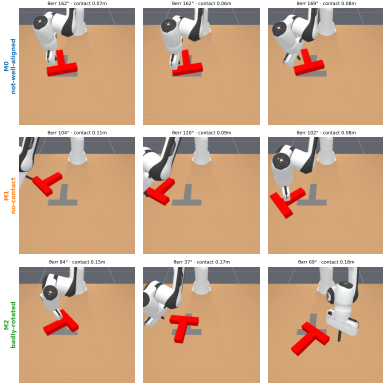


Figure 5: **Failure modes discovered on image-based Push-T: $k=3$ recurring modes** (*not-well-aligned*, *no-contact*, and *badly-rotated*), with three representative frames each, annotated with orientation error and contact distance. Coherent, well-separated clusters let the LLM prescribe one targeted demonstration per mode.

Push-T), $0.88/0.89$ (Lift), $0.82/0.86$ (Wipe), and $0.83/0.82$ (Door), a tight 0.82–0.89 range across all ten settings. A confident prescription is therefore evidence, available immediately, that the round will help. Low-confidence prescriptions ($< 60\%$) yield near-zero improvement while high-confidence ones ($\geq 90\%$) yield the largest gains, so a practitioner could audit or veto low-confidence prescriptions before expert effort is committed. Figure 5 shows the modes discovered on image-based Push-T: three coherent, well-separated clusters, internally consistent in orientation error and contact distance. These modes are the units over which the budget is allocated.

Conclusion

Interactive imitation learning has long asked *when* to query the expert. Under a fixed budget that is the wrong question: performance is set by *which* failures are corrected and *how* each demonstration is placed. DISTIL operationalizes this as demonstration distillation: flag uncertain rollouts with the learner’s own loss, root-cause failures with KAG-grounded foundation models, cluster them into modes with a cross-round memory, and prescribe each demonstration through a feasibility-checked LLM loop that corrects a chosen failure in place or bridges several related ones. Across five tasks, two observation modalities, and four learner architectures, the same loop attains the best mean success rate against six interactive baselines in every setting at a 20-demonstration budget, and its demonstrations carry the most new information, spread by design across distinct failure modes.

Limitations. On the robot tasks our experts are oracles (a privileged PPO policy and scripted motion planners); noisy or inconsistent human demonstrators would add variance this expert model does not capture (Mandlekar et al. 2020; Liu et al. 2023). Bridging placements require a resettable simulator, and all results are in simulation. The loop inherits the quality of its foundation models: weaker VLM or LLM

outputs degrade root-causing and prescription; KAG grounding, workspace clamps, and the feasibility loop bound, but do not eliminate, the damage of a bad prescription. Each round also pays the wall-clock cost of the reasoning pipeline (frame description, root-cause analysis, clustering, prescription, and possible re-prescription), an inference cost the per-state gates do not incur, and each new task family requires authoring its knowledge graph before the loop can run. The loop commits to a single demonstration per round and cannot exploit an expert willing to record several complementary corrections at once, and the descriptor ϕ_i and memory constants are designed, not learned. Letting the VLM’s failure taxonomy define the clustering space and extending prescription to human-gated hardware settings (Kelly et al. 2019) are natural next steps.

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